TMIRI

Davide Volpi

December 17, 2025

Contents

1 (Ir)rationality

We identify two cognitive modes in our brain:

- **Rationality:** believe in something by the identification of **relevant true premise**s and use them to reach sound **conclusions**.
- **Intuition:** believe something from **instinctive feeling** rather than conscious reasoning.

Doing research and/or innovation requires a good balance of **rationality and intuition**.

1.1 Rational Agent Assumption

- People are generally rational, but their thinking is normally sound.
- Emotions such as fear, affection, and hatred explain most of the occasion on which people depart from rationality.

The Homo Economicus Experiment: we always choose to perform the action with the optimal expected outcome for ourselves from among all feasible actions. The **Ultimate Game** shows if you are or not a Homo Economicus, where two people X and Y have to split $10 according to the following steps:

- X decides how the money is split.
- Y accepts the deal or rejects it. In case of rejection thet do not get anything.

1.2 Thinking Fast and Slow (Kahneman 2002)

States that we have two thinking modes:

- **Fast thinking:** intuitive, effortless, automatic, inaccurate
- **Slow thinking:** rational, tiring, self-awareness, more accurate

1.3 Cognitive Bias

A **cognitive bias** is a systematic pattern of deviation from rationality. Individuals create their own "subjective reality" from their perception of the input. An individual's construction of reality, not the objective input, may dictate their behaviour in the world.
Cognitive bias are many forms of **flawed perception, reasoning and decision making**. Are well identified cases of **predictable irrationality.** The reasoning errors associated to them are due to the design of our cognition machinery rather than the corruption of thought by emotion.
Examples: Confirmation, Overconfidence, Anchoring, Clustering, ...

1.3.1 Types of Cognitive Bias

Irrationality due to our Social Nature
The mere presence effect: not being alone changes our cognitive processes. Being with people makes us slightly smarter (if we do not feel pressure) or dumber (if we feel pressure).
Truth Telling Experiment (Abeler et al. 2012):

- Phone call asking to toss a coin.
- If heads 15 Euros immediately transferred.
- Almost all participants report their outcome honestly.

We can take advantage of our social irrationality by adding some elements to the contest.

Irrationality Assessing Likelihood of Events

- Dead causes by unnatural cases.
- Dead caused by a terrorist incident.

We are biased by social elements.

Irrationality Assessing Sensorial Feelings
Grading Pleasure Experiment: take two wines made out of the same type of grape. One is cheap and the other one is expensive:

- Grade them without knowing the price.
- Grade them knowing the price.
- Monitor enjoyment (brain sensors) without knowing the price (small difference).
- Monitor enjoyment (brain sensors) knowing the price (larger difference).

Irrationality Finding Patterns in the World
Clustering Illusion: during WWII, londoners developed complex theories about the pattern of bombs impact in London. But a statistical analysis shows that they were almost random.

Irrationality Finding Temporal Patterns
Hot Hand Phenomenon: you are the coach of a Basketball team. During a game, one second before the end, there is a tie. Your team has two free shots. Who should you assign the shooting?

- A player that has scored 10 free shots during the game without missing any.
- The player with the best free shots average last month.
- The player with the best free shots average along the reason.

Irrationality due to our Self-Confidence
The Dunning-Kruger Effect: people of low ability tend to overestimate their abilities.

Irrationality Revising our Beliefs
How politics makes us stupid (Kahan 2013):

- Researchers are experimenting with a new cream for skin rash. Considering the results of the experiment and please indicate whether the experiment shows that using the new cream is likely to make the skin condition better or worse. Results support that the better subjects were at math, the more likely they were to stop, work through the evidence, and find the right answer.
- Politicians are studying the effect of banning guns on crime rate. Consider the following data and please indicate whether the data indicates that banning guns is likely to increase or decrease crime rates. Results show that ideology drove the ability to find the right answer.

2 Critical Thinking

Is about coming out with opinions and knowing why we have them:

- Being the **owner of our opinions**.
- Being able to **hold our opinions in a conversation**.

Critical thinking is **hard**,**tiring** and it is not realistic to fully know the **reasons** of each one of our beliefs.
Negationism: extremist actors calling upon skepticism to negate scientific and historical real hypothesis.

$$Critical\ Thinking = Claim + Argumentation$$

2.1 Claim

Is an opinion, hypothesis, conclusion, solution that must be arguable (falsifiable) and precise (making falsification easier).

2.2 Argumentation

Is a support for the claim. Is formed by:

- **Premises:**
  - Evidence (observable facts).
  - Assumptions (beliefs generally accepted as true).
- **Reasoning:** link between evidence, assumptions and claim.

Argumentation can be:

- **Internal:** a thought process to convince ourselves.
- **External:** a public exposition to convince the others.

Ideally, arguments should be **irrefutable**. But this is not realistic in many contexts.
Argumentation Chains: formally, A and $A \to B_1, B_1 \to B_2, \ldots, B_{n-1} \to B_n$, therefore I claim B_n.
Hedging Language: words used to soften a claim, evidence, ... to make the argument more defensible.

2.2.1 Forms of Reasoning

Deduction
Moves from general statements to specific conclusions:

- **Deduction:** if a general rule is true, a particular case of it is also true.
- **Purpose:** deduction is used to infer specific cases.
- **Example:** *"All humans are mortal. I am human, therefore I am mortal"*
- **Soundness:** deduction is sound reasoning (if the premises are correct, we can be sure that also the conclusion is true).

Induction
Moves from specific cases to general rules:

- **Induction:** if many particular cases fit in a general rule, then the general rule is true.
- **Purpose:** induction is used to infer universal rules.
- **Example:** If I have seen many spiders and all them had 8 legs, I can make the general rule *"all spiders have 8 legs"*.
- **Unsoundness:** we cannot be sure of things obtained by induction.

Abduction

- **Abduction:** if there are several explanations for an observation, then the simplest explanation is the one that should be chosen (*Ockham's Razor*: fewest assumptions).
- **Purpose:** abduction is used to chose from different explanations of what we observe.
- **Example:** Ptolomeo's geocentric model vs Copernico's heliocentric model.

Forms of Reasoning: Example
Sven is a Swedish student that visited Barcelona as an Erasmus student last year. He tells his friend Olga that *Catalans are very friendly* (**induction**). He also says that the reason for that is the weather. *"Good weather makes people more social and therefore more friendly,"* (**adduction**) Sven says to Olga. Olga, who always trusts Sven, needs a caretaker for her mother and, obviously, wants her to be friendly. When she goes to a human resources agency to look for a candidate and *she sees that there is a girl from Barcelona, she immediately chooses her* (**deduction**). *See other examples in the slides*

2.2.2 Fallacies

Are a failure in reasoning which renders an argument invalid. They can be divided in:

- **Jumping from correlation to cause:** assuming that correlation implies causation.
- **Motte and Bailley:** someone proves a mild assertion (the motte) and erroneously asserts victory on a larger, more controversial question (the bailley).
- **Straw Man:** oversimplify a claim in order to make it less defensible.
- **Begging the Question:** when you use the point you're trying to prove as an argument to prove that very same point.It's also called circular reasoning.

2.3 Argumentation Spheres

Determine what is a valid claim, evidence and reasoning:

- **Personal:** actors are members of relationship, arguments are evaluated by group standards.
- **Technical:** actors are specialized members of the field, arguments are evaluated by field standards.
- **Public:** actors are journalists, politicians, ... arguments are evaluated by general public.

2.4 Habits of Critical Thinking

- **Disposition:**
  - **Curiosity:** desire to abroad the scope of your knowledge.
  - **Inquisitiveness:** actively desire to learn, pose questions, ...
  - **Open-Mindness:** being aware of your biases, empathy, ...
  - **Humbleness:** when finding contradiction to you beliefs.
  - **Self-Confidence:** making judgments, challenging others, ...
- **Metacognition:** being aware of your thinking.

2.5 Belief Updating

Imagine that you have some **belief** B with some **certainty** *C(B)*, you receive some **information** I for which you have **credibility** *C(I)*. Further, let *Cons(B,I)* represent the **consistency** between B and I:

- If *Cons(B,I)=high*, then you should **increase** *C(B)*.
- If *Cons(B,I)=low*, then you should **decrease** *C(B)*.

2.5.1 Bayesian Belief Updating

Bayes Theorem tells me how to update my believe as:

$$P(\mathcal{H}|\mathcal{D}) = \frac{P(\mathcal{D}|\mathcal{H}) \times P(\mathcal{H})}{P(\mathcal{D})}$$

2.6 How to Disagree

Ordered from the top to the basis of the pyramid:

1. **Gentrification:** move into poorer areas.
2. **Refusing the central point**
3. **Refutation:** find reasoning mistakes.
4. **Counterargument**
5. **Contradiction**
6. **Responding to tone**
7. **Ad hominem:** attacks the characteristics without addressing the substance of the argument.
8. **Name calling:** using insults/negative labels.

2.7 Case Study

2.7.1 Educational Level of Barcelona Population

- **Observation from Data:** people from the poorest neighbourhoods are 10 times less likely yo obtain a university degree than people from the richest neighbourhoods.
- **Social Determinism:** It is a Political Sciences theory that claims that your life is highly determined by your social strata.

The data seems to support Social Determinism. Therefore, accepting the observation one should increase the certainty about Social Determinism.
Fallacy: Jumping from correlation to cause, claiming this theory after seeing the data.

2.7.2 The Untold Story Behind DNA Similarity

Analyzing the Argumentation: in a well-written paper, it is easy to identify claims, assumptions, facts, reasoning chains, ...

- **Argumentation Sphere:** this is a popular paper, so biologist should go easy on its soundness and casual readers should give some trust to the authors.
- **Assumption:**
  - **Claim:** humans and chimpanzees have a common ancestor.
  - **Argumentation:** this is indisputable because their DNS is similar up to 98%.
- **Paper's Claim:** the generally accepted claim needs to be revisited because
- **Argumentation:**
  - The 98% of the DNA similarity is not true
    * Assembling huge chromosomes from small snippets is very challenging (new research 85%).
    * Contamination from human DNA
    * Researchers are biased. Bagging the question fallacy.
  - Similarity does not imply common ancestor
    * The paper does not develop this idea, buy I can buy it.

Fallacy: straw man, can be seen from the claim the assumption, that is oversimplified.
Real Accepted Claim: evolution is believed to be true because of a huge amount of evidence in genetics, geology, biology. One of the many claims fitting the evolution theory is that humans and chimpanzees have common ancestor and again this is true because of a huge amount of evidence in genetics and anthropology.

2.7.3 California has a Climate Problem, and its Name is Cars

Claim: decarbonizing the electricity sector is both possible and profitable. To reach the ambitious carbon targets states will have to decarbonize transportation.
The paper questions the previous accepted claim:

- **Accepted Claim:** reducing air pollution → hurts economical growth.
- **Claims (with chain of claims):**
  1. Reducing air pollution → helps economical growth.
     - **Argumentation:**
       * GPD per capita has grown double than national average while dropping more emissions.
       * Employment bonanza.
       * Innovation bonanza.
       * Energy productive economy
- **Problem:** this works only for California due to the presence of more green implants and the silicon valley, plus a lot of other stuff that make this just a coincidence and not a true statement.

3 Epistemology is a Nutshell

Epistemology: is the branch of philosophy that examinates the nature, origin, and limits of knowledge.
Science: is about discovering new true knowledge and making it available to the world.

3.1 Some Milestones and Keywords in the History of Science

Greek Scholars, 400 BC: they were looking for natural explanation of phenomena, thinking about what is to think, and they became aware of our cognitive processes.

- **Plato:** we cannot trust observation. "Discovered" **rationalism**, knowledge can only be gained by reasoning.
- **Aristotle:** premises are learned from observation. "Discovered" **empiricism**, knowledge has to be gained by observation. Is the first formalization of **reasoning and deductive reasoning**.

Enlightment (XVII): Sapere Aude (dare to think by yourself). Mankind instead of God is at the center of the universe, have the courage to use your own understanding.

- **Galileo:** Separate Science from Theology. If empirical evidence contradicts religion, religion cannot censor it. Introduced the **Systematic Observation**, systematically record data about the phenomenon under study before assuming any thing, stating implicitly the **inductive reasoning**.

Skepticism (1700): Hume with **problem of induction**, we never know if the next observation will satisfy the claim.
Idealism (1830): Focus on the Fundamental Nature of Being (too theoretical and useless in practice).
Modern Science (1900):

- **Logical Positivism** (Wien Circle):
  - **Meaningful:** it must be possible to determine the truth (**verification principle**).

Karl Popper (Wien 1902):

- **Logical Empiricism:** a statement is useful if it is **falsifiable**.
  - Replace verification principle by confirmation.
  - Science is about statements that can be confirmed or falsified.
- **Consecutive Empiricism:**
  - Science attempts to construct theories that explain empirical results.
  - Scientific claims are taken as true only as far as the observations go along.
  - Hypothetic-deductive Scientific Method.

3.2 Scientific Method

1. **Observe** and try to make sense out of what you observe.
2. **Inductive step:** from observation and thoughts come out with a hypothesis.
3. **Deductive step/experimental design:** make a prediction that should happen if the hypothesis is true.
4. **Testing step/experiment:** check if the prediction came true.
5. **Evaluation:** if prediction/experiment outcome goes along hypothesis, then increase likelihood of hypothesis. Else reverse or reject hypothesis.

Example: The Titus-Bode Law
J.E. Bode claimed that the distance of planets wrt sun follows a geometric progression of power 2. The law, which fitted the known planets, was considered interesting. It gained popularity when it rightly predicted Uranus and Ceres. It was widely accepted at that point, until in 1846 Neptune was discovered in a location that does not conform to the law. The discovery of Pluto in 1930 confounded the issue still further. Now it is considered a spurious curiosity.

3.2.1 Verification vs Falsification

- **Verification (Failed Falsification):** this involves seeking evidence to confirm a theory.
- **Falsification:** this involves seeking evidence that disproves a theory.

Popper: our confidence in a theory isn't based on how many times it's been proven right, but on how many times it has been rigorously tested and failed to be proven wrong.

3.3 Scientific Attitude

Science is a **Social Activity** and scientists must be:

- **Open:** willing publicise work.
- **Critical:** willing to give and get constructive feedback.
- **Transparent:** willing to allow replicability.

Example of Success: Nobel prize in Economics 2021
The prize was awarded to David Card, Joshua Angrist and Guido Imbens for real-word research in the 1990s that demonstrated, empirically, that the idea touted by conservative economist that higher minimum wages mean fewer jobs is not based on fact.

Example of Failure: Michael Siegel, public-health researcher at Tufts
Most of the research which has been funded by alcohol companies has reported that there is a benefit to drinking moderate amounts of alcohol. Most of the research that has not been funded by the industry has found that there actually is not a benefit, and that overall mortality is higher.

3.4 Scientific Method and Attitude

R. Feynman: science is what we do to avoid lying to ourselves.
That's what distinguishes science from **pseudo-science or negationism:**

- Pseudo-scientist see what they belied, instead of believing what they see.
- A decision is made beforehand about what one wants to be true.
- Then high acceptance standards are required for the things that they do not want to believe and very low for the things that they want to believe.

4 Types of Research

Demarcation Problem: in philosophy of science and epistemology, is the question of how to distinguish between science and non science. It also examines the boundaries between science, pseudoscience and other products of human activity, like art and literature.
General Definition: research is any topic/activity in which one can make a claim about a topic of interest and defend it with a rational argumentation.

- **Theoretical:** structured approach, applied to an abstraction of the real world. The purpose is to prove statements and the analysis/argumentation is based on definitions and formal logic. More common in mathematics.
- **Quantitative:** structured approach with the purpose to quantify something. The analysis/argumentation is typically based on data and statistics. More common in natural science.
- **Qualitative:** flexible approach, with the purpose of describe/explain in a meaningful/convincing way. The analysis/argumentation is typically based on meaningful examples and a meaningful narrative. More common in social science.

4.1 Quantitative Research

- **Longitudinal:** data is collected for a group of individuals over prolonged periods of time.
  - May be confounded by temporal features.
- **Cross-sectional:** data is collected for a group of individuals without considering time (snapshot).
  - May be confounded by inter-personal features.

4.1.1 How we Observe

Let's suppose we want to see the relation between an independent variable has on a dependent variable:

- **Experimental:** we make an intervention in the data to study its effect (may be impossible/unethical/complicated). We can intervene on the independent variable. **Cause-relation** can, in general, be proved.
  - **External Validity:** some times we cannot guarantee that the sample we use for the experiment is representative of the total population.
  - **Random Control Trials(RCT):**
    1. Find a representative sample of the population under study.
    2. Assign randomly subjects to the different groups.
    3. Because of randomness, the different groups are statistically equivalent wrt every known or unknown variable.
    4. Manipulate the independent variable in each different group to determine its effect in the dependent variable.
- **Observational:** we do not make an intervention in the data (just observe). We cannot intervene on the independent variable. **Correlation** can, in general, be proved. Cause-relation can, in general, only be **conjectured** and based on assumptions about the domain (**conjectured causality**).
  - **Descriptive:** Consider **one variable** of interest and the variable is not manipulated. Seeks to **describe** its status and is designed to provide **systematic information** about a phenomenon. The researcher does not usually begin with a hypothesis, but is likely to develop one after collecting data.
  - **Correlational:** Considers **two or more variables** of interest that are **not manipulated**. Determines the extent of their **relationship** using statistical data. Does not prove the cause of the pattern and cannot be sure if they are spurious.
    The scientist may use it to see correlation as a first step before understanding the connection.

- **Causal-comparative:** Considers two or more variables that are not manipulated and **attempts to establish cause-effect relationships among the variables**. The strength of the experiment will depend on **how close to randomness** nature seemed to behave. Determine causality must be done carefully, as other variables, both known and unknown, could still affect the outcome.
  The researcher cannot randomly assign groups and must use ones that are naturally formed.

4.2 Types of Research

- **Basic Research:** understanding the world for the sake of curiosity.
- **Applied Research:** discovering how to take advantage of our understanding of the world.
- **Development and Innovation:** making applied research readily available.

4.3 Design Science Research (DSR)

Quantitative and qualitative research in the creation of artifacts and systems, and their embedding in our physical, virtual, psychological, ecnomic, and social environment.

- Standard vs Design Science:
  - **Standard Science:** explains an existing reality, or helps to make sense of it.
  - **Design Science:** construct a new reality.

Design Science refers to a scientific approach to design solutions.
DSR can be seen as the Scientific Method for Applied Research.
It is done as follows:

1. **Identification** of the problem and **justifying** the value of a solution.
2. **Definition** of objectives for a solution.
3. **Design**
4. **Development** of the modules/components.
5. **Demonstration** by using the artifact to solve the problem.
6. **Evaluation** of the solution, comparing the objectives and the actual observed results.

4.4 Research in Computer Science

A rough classification of research in CS is done as:

- **Formal or Theoretical:**
  - **Prove** properties about paradigms, algorithms, ... by capturing the fundaments of the problem, identifying fundamental possibilities and limitations.
  - Mainly uses **deduction** and requires a **formal description**.
- **Build:** building an artifact.
  - The DSR methodology applies.
  - Is concerned with **proof of concepts**, **novelty**, **good design** with documentation and **good evaluation** with benchmarks/report of measures.
- **Process:** understand/improve the processes used to accomplish tasks in which we use computers.
  - **Method:** standard scientific method if the goal is to understand or DSR if the goal is to improve.
  - Concerned with computers as a tool/entertainment device, human behaviour. It needs to deal with imprecise concepts and is essential to achieve the maximum scientific rigor possible (hypothesis, experiments, results).
  - Is usually quantitative research, but sometimes it is qualitative.

- **Model:** defining an abstract model for a real system and use computers to answer useful questions.
  - Concerned with computers as computing machines, are less complex than the real system while preserving fundamental aspects.
  - Is a simulation (quantitative research) that can verify properties on the model (theoretical research).

5 Ethics for Scientist and Engineers

Harmless Stories (J. Haidt): where moral values come from?

- Hypothesize that morality is a **human construct** and **pragmatical construct** (evolution-driven).

5.1 Morality vs Ethics

- **Morality:** collective agreement about what is right and wrong (shared principles).
- **Ethics:** field of philosophy about addressing morality in a critical objective way.

5.2 Morality

- **Morality and Religion:** along history, religion has strongly influenced morality and priests have helped people to deal with it. However, the argumentation has often been weak.
- **Morality without Religion:** moral principles are behaviour guidelines/principles that we all voluntarily follow because it makes the world a better place for all of us.

In some cases morality follows **universal principles** (e.g. killing, but has some "exceptions"), but in many other cases it is strongly based on context (e.g. sexism, slavery, ...).

5.2.1 Judging Morality

We have general moral principles, and sometimes we see that somebody, in a particular context, breaks them; then we make a moral judgment. There are two very tempting extreme approaches:

- **Relativism:** all moral points of view are equally valid.
  - Right/wrong for one person is not necessarily so for another, depends on time/place/culture.
  - **Problems:** invalidates moral judging and makes moral discussion impossible and meaningless.
- **Absolutism:** there is a system of norms and values universally applicable.
  - **No exception:** a rule is a rule.
  - **Problems:** invalidates moral judgment, hides the complexity of many dilemmas and leaves no room for independent moral judgment.

A very useful and common argumentation for moral judgments is **Argumentation by Analogy:** if something is clearly ethical/unethical in an example case, then it is also so for a comparable situation.

5.3 Ethics

Ethics helps us address morality in a rational way such as moral judgments in complex situations or decide the right course of action. A first ethical consideration where it is not the same:

- **Must do:** help a person in need near you.
- **May do:** donate to a charity.

This is important, because people violating those rules deserve different moral judgments.

Example of Ethics vs Law: Hammurabi Code

- If a man accuses another man and charges him with homicide, but cannot bring proof against him, his accuser shall be killed.
- If a builder constructs a house for a man but does not make it conform to specifications so that a wall then buckles, that builder shall make that wall sound using his own money.

Both want to encode rules for a civilized society:

- **Law:** must be precise as possible (difficult for changing environments) and must be followed.
- **Ethics:** personal interpretation. Social pressure and self-esteem is what may enforce morality.

If you do not care about ethics, it means that you embrace the moral beliefs and norms given to you by your family and your society. You accept them without question or serious examination. In other words: you do not think critically about your moral principles.
Grouch Marx: those are my moral principles, and if you don't like them, well, I have others.

- We align our values to our incentives.
- It seems a good idea to anchor our moral values as much as possible before we have a moral conflict.

5.3.1 Ethical Dilemmas / Thought Experiments

Ethical Dilemma: solution in which a difficult choice has to be made between two courses of action either of which entails transgression of a moral principle.
Thought Experiments: hypothetical situation in which a hypothesis, theory, or principle is laid out for the purpose of thinking through its consequences.

5.4 Ethical Theories

Structure of Human Action: an action is carried out by a certain actor which has consequences.

- **Values:** lasting convictions about the society that we all want.
- **Norms:** rules that tell what actions are required, permitted or forbidden.
- **Virtues:** desirable human quality.

We can **classify ethical theories** as:

- **Consequentialism:** look at consequences-values.
- **Deontology:** look at actions-norms.
- **Virtue Ethics:** look at actors-virtuousisy.

5.4.1 Utilitarism (Jeremy Bentham 1748)

- Actions are **judged** by the amount of **pleasure and pain** they bring about.
- People deserve to be treated ethically because they are **sentient**.
- **Utility:** chose the action that brings the greatest happiness for the greatest number of people.

Is a **consequentialist** theory where happiness/well-being is the value of interest.
The **assumption** is that pleasure and pain can be measured. This assumption is not realistic, problematic in practice and an extreme trust in this theory is "the end justifies the means".
In many situations the computation of happiness has to be adjusted.

5.4.2 Kantian Theory (1724)

- We should be able to determine **norms** telling what is right/wrong through **reasoning**.
- Actions are judged from how they fit those norms and we should follow the norms of **duty**.
- People deserves to be treated ethically because they are **rational beings**.

It is a **deontological** theory where ethics is about moral rules.
The **assumptions** are that is possible to make a universal deontological code and trust in human kind.
Kant claims that **ethical guidance** can be identified with the following two principles:

- **Universality:** act as you would like everybody else to act in similar situations.
- **Reciprocity:** treat others as an end, not as a mean.

Motivation is duty: not look after our sick mother out of love or compassion, . . . but out of duty.
The cope with this problem is that norms are arranged as a hierarchy. Rules are to be followed unless it is inconsistent with a more important rule.

5.4.3 Virtue Ethics (Aristotle 384)

- There are some **personality traits** (virtues) that define **morally good people**.
- When confronted with a moral dilemma ask *"what should I be?"*
- Virtuous people live a good life, meaningful and that makes them happy (special type of happiness).
- An action is **morally acceptable** if that action is what a virtuous person would do.

Assumption: we can identify or imagine virtuous people and be inspired by them.
Virtues can be classified:

- **Intellectual Virtues:** wisdom, rationality, ...
- **Moral Virtues:** fairness, benevolence, honesty, ...
- **Professional Virtues:** will depend on the profession.

Golden Mean: virtues require a balance between two behavioural extremes.

	Utilitarism	**Kantian**	**Virtue**
Look at	Conseq. confronted w. values (happin)	Actions confronted with norms	Actors confronted with virtues
Why	"victims" are sentient	"victims" are rational	we want to be virtuous
Tool	Moral Balance Sheet	Universality and Recipr. Principles	Virtuous People
Motiv.	Max happiness	Duty	Good life

6 Ethics in Research

6.1 Malpractice

6.1.1 Fraud

Publication of knowledge based on false evidence.

- **Consequences:** contamination of literature and social loss of trust in the community.
- **Solution:** make it harder by requesting transparency and funding for replication studies.

6.1.2 Plagiarism

Appropriation of other's work without due credit (text, data, ideas).

- **Consequences:** theft of intellectual rights and intellectual credits.
- **Solution:** transparency, automatic tools, publish in reputable journals and conferences.

6.1.3 Conflict of Interests

Researcher has other external interests that may clash with the claims made in the research.

- **Consequences:** contamination of literature.
- **Solution:** what is wrong it not to disclose the conflict, just be transparent (acknowledgments).

6.1.4 Undue Influence of Personal Values

Researcher is biased towards proving certain claims (ideological bias).

- **Solution:** difficult to solve, you cannot ask researchers to disclosure their personal values.

6.2 Questionable Research Practice

6.2.1 Data Massage

Selective report of results (sometimes is legitimate)

- **Solution:** report the manipulation and if possible, pre-decide/register the manipulation rules.

6.2.2 P-Hacking

Manipulate results to reach the right confidence level of the statistical test.

6.2.3 Cherry Picking

Reporting only favourable results.

- **Solution:** tell the truth (idea may work well on some relevant instances).

6.2.4 Data Snooping

Stop the data collection when results are favourable.

- **Solution:** tell the truth (with longer cutouts the advantage of the algorithm does not show clearly).

6.2.5 HARKing

Hypothesize After Results are Known: presenting a post hoc hypothesis in the introduction of a research report as if it were an a priori hypothesis.

- **Legitimate:** run experiments to observe (induction phase).
- **Illegitimate:** present the data as if obtained in the deduction phase.

7 Bioethics

Ethics applied to life science. The **Fundamental Principles of Bioethics** are:

- **Do Not Harm**: obligation to not harm the subjects.
- **Do Good(Beneficence):** obligation to act for the benefit of society.
- **Autonomy:** subjects have the power to make rational decisions and moral choices, and each should be allowed to exercise his or her capacity for self-determination.
- **Fairness (Justice):** fair, equitable, and appropriate treatment of persons.

Conflicts Between Principles: Each one of the principles of ethics is to be taken as a *prima facie* obligation that must be fulfilled, unless it conflicts, in a specific instance, with another principle. When faced with such a conflict, the researcher has to determine the actual obligation of the subjects by examining the respective weights of the competing obligations based on both content and context.
The influence of the four principles can be seen in the protocols for testing and marketing new drugs.

7.1 Animalist Movements

Give the same moral value to animals and humans because animals are **sentients** and**intelligent**.
Two points of view of two prominent and influent philosophers:

- **Pete Singer (utilitarian):** feelings of animals have the same value as of humans.
- **Tom Regan (deontological):** animals have the same right to be treated with respect and dignity.

7.1.1 Utilitarian Approach

Moral balance sheet between good consequences (hedons) and bad consequences (dolos).
How we quantify pleasure and pain must be weighted according to the answer to some questions.
Important to decide if we consider animals suffering of less importance as we go down in the sophistication.

7.1.2 Kanthian Approach

If we consider all animals deserving the same moral value (if animals deserve respect):

- **Universality principle**
- **Reciprocity principle:** treat animals as an end, not as a mean.

7.1.3 Virtue Ehics

Humans typical mistreatment of animals, exhibits serious vices of character. We appreciate people who exhibit virtues as kindness and compassion in their treatment of animals. Therefore, a virtuous person cannot ignore the mistreatment of animals.

7.1.4 A Pragmatic Trade-Off

A rule of thumb to judge the morality of a particular experiment with animals is to confront it with the **Principle of the three R's**:

- Replace (simpler animals)
- Reduce (less animals)
- Refine (less pain)

An independent ethical board must approve the research.

7.2 Independent Ethical Committees

Those conducting the research may be biased about the research design so there are **Ethical Committees** that have an independent board approving and supervising the study.

7.3 Research with Humans

Informed Consent: if any research is conducted on a human subject, then this human subject must be informed about the experiment, must consent to the experiment voluntarily, without any coercion, and must have the right to withdraw consent at any time.

- Sometimes complete information is not possible.
- Informed consent is required if it is research, but is not required if it is the ordinary conduct.
- Ethical issues about asking for consent such as when, how and what is asked.

7.4 Study Case

7.4.1 Neuroscience

- **Study:** placing electrodes on inmates foreheads and sending a current into their brains. The electricity will target the prefrontal cortex (decision making and social behaviour). The idea to be tested is that stimulating more activity in that region may make the prisoners less aggressive.
- **In favor:** many potential future benefits.
- **Against:** consent is given in a coercive situation.

7.4.2 Journal of Business Research

- **Study:** predicting poker players' churn in the online gambling industry using real-world data from bwin. The work benchmarks single models against ensemble methods to identify gamblers likely to leave.
- **In favor:** churn prediction helps efficiently retain customers, improves CRM strategies, and provides robust identification and profiling of at-risk players.
- **Against:** potential ethical concerns about targeting vulnerable gamblers, plus model bias and reliance on behavioral data that may not capture the full context of churn.

8 Ethics for Engineers

8.1 Responsability

Progress has risk, should we stop progress? From a utilitarian perspective, a small risk is clearly outweighted by the long-term benefits. From a deontological prospective, a small risk may be accepted as long as decisions have been made thoughtfully and risk have been minimized and are very small.
Liability: even though we can accept that progress has risks, if something goes wrong, it is imperative to know if there is somebody to be blamed.
The problem of Many Hands: when many people are involved in an activity, it is often difficult, if not impossible, to pinpoint who is morally responsible for what.
When work is done by teams, it should be clear who is responsible for what.
Responsible Engineers: we want to guarantee that nothing goes wrong and, if it does, we want to feel that our behaviour was correct.

8.1.1 Passive Responsibility

Whenever something goes wrong the question who is responsible for it often quickly arises. Usually four conditions need to apply for an actor to be held responsible:

- **Wrong-doing:** a moral norm has been violated or a value has been threaten.
- **Causal Contribution:** the consequences of the action must be reasonably related to the harm.
- **Foreseeability:** the harmful consequences of the action should be reasonably predictable.
- **Freedom of action:** an alternative course of action was available.

8.1.2 Active Responsibility

Act taking into account that you do not want to be blamed if anything goes wrong.

- Engineers are usually hierarchically below managers and sometimes they disagree.
- How to deal with ethical conflicts?
  - **Separatism:** engineers are responsible for the design or development and not for consequences.
  - **Technocracy:** let engineers decide.
- **Cooperation:** a company should have dialogue channels where engineers and managers exchange their perspectives.

Whistle-Blowing: disclosure of abuses in a company by an employee, without the consent of his superiors. It violates deontological rules and professional values.
When is whistle-blowing morally required?

- The company does serious harm to the public.
- The would-be whistle-blower has identified the harm, reported it to the superiors, and concluded that they will do nothing.
- The would-be whistle-blower has exhausted other internal procedures within the company.
- The would-be whistle-blower has evidence that would convince a reasonable, impartial observer.
- The would-be whistle-blower has good reason to believe that revealing the information will prevent or decrease the harm.

8.1.3 Codes of Conduct (Deontological Code)

Guidelines given by organizations for responsible behaviour of their members. There are two types:

- **Professional:** integrity, honesty, social responsibility, ...
- **Corporate:** mission statement, core values, norms and rules, ...

The **ACM Code of Ethics and Professional Conduct** contains:

1. Fundamental ethical considerations
   - **Contribute** to the society and human well being.
   - **Avoid harm** to others.
   - **Be honest** and trustworthy.
   - **Be fair** and take action not to discriminate.
   - Honor **property rights**.
   - Respect **privacy**
2. Specific ethical considerations
3. For people in leadership roles
4. Principles involving compliance with the code

8.2 Humane Decision making and Innovation

8.2.1 The Ethical Cycle

A methodological tool to structure our analysis on how to deal with Professional Ethical Dilemmas.

1. Moral precise problem statement (what is the problem, who has to decide, the moral nature).
2. Problem analysis (relevant data, i.e. stakeholders, facts, moral values, norms).
3. Options for action (identify alternatives, from the obvious ones to the creative ones).
4. Ethical evaluation (analyze alternatives with intuition, common sense, by analogy, with principles).
5. Reflection (compare alternatives with theories).
6. Moral acceptable action (chose alternative)

8.2.2 Humane Technology

Design and develop technology in a way that supports our well-being, democratic functioning, and shared information environment.

8.2.3 The Three Laws of Humane Tech

Proposed by Center for Humane Technology with the purpose to simplify what does it mean to be a responsible technologist. The motivation was that Aza Raskin in 2006 invented the infinite scroll:

- Invented in one context and adopted by social media.
- **Effect:** 200.000 human lifetimes per day being wasted.
- His mindset in 2007 was "infinite scroll is cool, so let's do it", but now is "sorry world!"

The three created laws are:

1. When you invent a new technology, you uncover a new class of responsibilities. And it's not always obvious what those responsibilities are.
2. If that invention, confers power, it will start a race.
3. If you do not coordinate and set norms, that race will end in tragedy.

8.2.4 Humane Design

Design decisions have social consequences, need to be unveiled and analyzed (ethics must be considered).
Moral Design: the deliberate design of a product aiming at users acting ethically.

8.3 Ethics and AI

The main risk of AI is **Human Stupidity**: humans have cognitive vulnerability, but AI is very persuasive (designed to convince us) and sicofant (designed to pleasure us).

9 Scientific Ecosystem & Metrics

9.1 Journals vs. Conferences

Research is a social activity requiring critical discussion.

- **Conferences:**
  - *Goal:* Dissemination and feedback.
  - *Process:* Fast turnaround (1-2 months), usually 0 or 1 review iteration.
  - *Types:* **General** (huge, networking) vs. **Specific** (small, socializing).
  - *Warning:* Beware of "Fake Conferences" (pay for a vacation).
- **Journals:**
  - *Goal:* Reliable source of specialized knowledge; archival.
  - *Process:* Slow (months to years), unbounded review iterations.
  - *Structure:* Editor-in-Chief (ultimate responsibility) → Area Editors (handle process) → Reviewers (assess quality).

9.2 Publishing Models

- **Open Access:** Free online availability. Solves the problem of rising subscription fees for libraries.
- **Predatory Journals:** They charge publication fees ($500-$2000) but provide no editorial services or rigorous peer review.

9.3 Quantitative Metrics

- **Impact Factor (IF):**
  - Average citations in year Y to papers published in $Y-1$ and $Y-2$.
  - *Flaws:* Arithmetic mean (skewed by outliers), unstable, manipulatable by editors.
- **H-index:** An author has index h if they have h papers cited at least h times.
- **DORA (San Francisco Declaration):**
  - Do **not** use journal-based metrics (like IF) to assess individual researchers.
  - Focus on content and qualitative indicators (influence on policy/practice).
- **Goodhart's Law:** "When a measure becomes a target, it ceases to be a good measure."

10 The Peer Review Process

10.1 Roles and Responsibilities

- **Authors:** Responsible for ethical standards and careful preparation.
- **Referees:** Must be fair, objective, and accurate. Should only recommend acceptance if confident.
- **Editors:** Select referees, supervise the process, and handle arbitrage.

10.2 Evaluation Criteria (The "Contribution")

A paper must contain a contribution = **Originality + Interest + Validity**.

- **Originality:** Are ideas new? Ranges from incremental to groundbreaking.
- **Interest:** Significance (Who cares?) and Relevance. "Obvious" ideas can still be interesting if they provide new insight.
- **Validity:** Soundness of the Scientific Method.

10.3 Biases and Problems

- **The Referee's Dilemma:** False Positive (accepting flawed work) vs. False Negative (rejecting good work). *Strategy:* In doubt, suggest rejection.
- **Decision Fatigue:** Reviewers get tired. *Advice:* Review in the morning.
- **Bohannon's Experiment:** 60% of fake papers were accepted by fee-charging journals.

11 Case Study - The Peter Principle

11.1 The Theory

- **Peter Principle:** "Every new member climbs the hierarchy until he/she reaches his/her level of maximum incompetence".
- **Common Sense (CS) Hypothesis:** Competence is preserved ($C_{new} \approx C_{old}$).
- **Peter Hypothesis (PH):** Competence at new level is random/independent.

11.2 Results of Promotion Strategies

- **Promote The Best:** If PH holds, efficiency collapses (-10%). Agents rise until they hit their "min" competence.
- **Promote The Worst:** If PH holds, efficiency increases (+12%). Agents rise until they hit their "max" competence.
- **Random Promotion:** Works moderately well regardless of hypothesis.

12 Technical Writing: Math & English

12.1 General Rules & Styles

- **Plain Text:** Never use plain text for math (e.g., don't write "n" for a variable, use n).
- **Styles:**
  - Scalars: Italics (x, y, a, b).
  - Vectors: Bold or arrow ($\mathbf{v}, \vec{v}$).
  - Sets: Uppercase (S, T).
  - Functions: Standard functions (sin, max, log) in Roman type; custom functions (f, g) in italics.
- **Notational Entity:** Use "heavier" styles for more complex objects.

$$\text{Calligraphic } (\mathcal{P}) > \text{Uppercase } (P) > \text{Lowercase } (p)$$

- **Coherence:** Do not use the same letter in different styles unless strictly related (e.g., $c \in C$). Avoid $c \in \mathcal{C}$ as they look too similar.

12.2 Definitions

- **Simplicity vs. Efficiency:** Definitions are not algorithms. When defining a set, prefer the form that is easiest to grasp, not the one that is computationally efficient.

$$\text{Better: } \{a \in \mathbb{N} \mid \exists b, b^2 = a\} \quad \text{Worse: } \{a \in \mathbb{N} \mid \exists_{0<b\leqslant a/2} b^2 = a\}$$

- **Case-based Definitions:** Conditions must be mutually exclusive and cover all cases (use "otherwise" for the last case).
- **Modularity:** Break complex definitions into simpler concepts.
- **Naming Elements:** Do not name the elements of a set in the definition (e.g., "Let $X = \{x_1 ... x_n\}$") unless you need to refer to specific indices later. Just say "Let X be a set."

12.3 Knuth's Hints

- Facilitate memorization (p for point, s for set).
- Avoid unnecessary subscripts (x and y is better than x_1 and x_2).
- Avoid piling subscripts (e.g., x_{i_j}).
- Avoid unfamiliar symbols or accents.

13 Writing Algorithms

13.1 Concept

Writing an algorithm is different from implementing it.

- **Goal:** Communication and clarity, not efficiency.
- **Process:** Done *a posteriori* (after coding). It is "coding your thoughts."
- **Advice:** Avoid implementation bias. Do not show variable initializations (e.g., $i = 0$) unless crucial to the logic.

13.2 Examples of Good vs. Bad Formalization

- **The Celebrity Problem:**
  - *Naive:* Check everyone against everyone ($O(n^2)$).
  - *Formal Property:* If u knows v, u is not a celebrity. If u doesn't know v, v is not a celebrity.
  - *Efficient Algorithm:* Maintain a "Candidate" set. Pick two, eliminate one based on the property. $O(n)$.
- **N-Queens (Constraint Formalization):** Instead of just saying "queens don't attack," formalize the variables:
  - Variables $x_1 \dots x_n$ (rows) with values $1 \dots n$ (columns).
  - Constraints: $x_i \neq x_j$ (different columns) AND $|x_i - x_j| \neq |i - j|$ (different diagonals).
- **Round Robin:** Define it mathematically as a matrix $M_{n \times n-1}$ where $M(i, j)$ is the opponent of team i at week j.

13.3 Procedural vs. Declarative Definitions

Usually, declarative definitions are better. However, sometimes a concept is best defined by the procedure to create it.

- *Example:* **Induced Width** of a graph. It is defined by the process of eliminating nodes in order and connecting their neighbors. Hard to define without the "procedure."

14 Structure of a Scientific Paper

14.1 Core Philosophy

- **Formula:** Paper = Claim (Hypothesis) + Argumentation (Support).
- **Criteria:** It must demonstrate Relevance, Originality, and Validity.
- **Composition:**
  - It is **not a novel**: No secrets, reveal the ending immediately.
  - It is **not a diary**: Do not write about your struggle or the chronological path you took. Write the logical path.

14.2 Selective Reading & Section Breakdown

Most readers do not read the whole paper. You must write for different levels of attention.

1. **Title:** Sexy, concise, informative. (Reads: $\sim$300)
2. **Abstract (The "Trailer"):** (Reads: $\sim$100)
   - A snapshot of the paper.
   - Self-contained, no acronyms, no math.
   - Summary of scope, achievement, and conclusions.
3. **Introduction (The "Plot"):** (Reads: $\sim$50)
   - Expanded abstract. Explain the problem, the approach, and *why* it is original/useful.
4. **Preliminaries:** (Reads: $\sim$10)
   - Background knowledge. Be concise. Cite sources but adapt notation to your needs.
5. **Contribution (The "Core"):** (Reads: $\sim$20)
   - Logical chain of reasoning supporting the claim.
   - Structures: Chain (A leads to B), By Specificity (Outline then details), By Example (Chess then General), By Complexity (Integers then Rings).
6. **Related Work:**
   - Puts your work in context. Can be placed after Intro, before Conclusions, or diluted throughout.
7. **Conclusions:**
   - Do **not** repeat the abstract.
   - Speculate on extensions and consequences.

14.3 Writing Process

1. Write a template/outline.
2. Sketch sections (20-200 words).
3. Draft the Contribution (the meat).
4. Iterate/Refine.
5. Write Abstract and Conclusions **last**.